

RODRIGO JOSE GONZALEZ HERNANDEZ

Oden Institute for Computational Engineering & Sciences
University of Texas at Austin
Texas, United States.

+(1) 404-542-0023
rodrigogonzalez@utexas.edu
Mailing Address: 201 E 24th St, POB 3.414

Education

University of Texas at Austin

Austin, Texas, United States

Ph.D. Candidate in Computational Science, Engineering, and Mathematics

2022-Present

- **Advisor:** Irene M. Gamba
- **Co-Advisor:** Diego Del-Castillo-Negrete
- **GPA:** 3.9/4.0

Emory University

Atlanta, Georgia, United States

B.S. in Applied Mathematics

2018-2022

- **Thesis:** Application of the Double Descent Color Intermittent Diffusion Method to a Variational Data Assimilation Problem
- **Advisors:** Manuela Manetta, Alessandro Veneziani
- **Major GPA:** 4.0/4.0

Research Interests

- **Active interests:** numerical analysis, scientific computing, discontinuous Galerkin finite element methods, spectral methods, low-rank and structure-preserving methods for kinetic equations.

Research, preprints and publications

In Preparation:

- *A structure-preserving local discontinuous Galerkin method for the Fokker-Planck-Landau equation.* Joint work with Kun Huang, Andres Galindo, and Irene Gamba.
- *Galerkin Petrov Approach for the Landau equation.* Joint work with Irene Gamba.
- *Galerkin Petrov Approach for Relativistic Collisional Models.* Joint work with Irene Gamba, Kun Huang, and Chi-Wang Shu.

Invited Talks and Presentations

- *Unconditionally conservative local discontinuous Galerkin method for the non-linear Fokker-Planck-Landau equation.* 33rd International Symposium on Rarefied Gas Dynamics, Göttingen, Germany, July 2024.
- *Numerical stability of weak forms.* ODEN Institute student seminar, Austin, Texas, USA, April 2024.

Distinctions

- Stephen Della Pietra Graduate Fellowship, Simons Laufer Mathematical Sciences Institute, Berkeley, CA. August 2025 - December 2025
- National Initiative for Modeling and Simulation (NIMS) Graduate Research Fellowship , The University of Texas at Austin, 2022-Currently.
- Trevor Evans award for outstanding undergraduate majors in the Department of Mathematics at Emory University, 2022.
- Phi Beta Kappa Honor Society member, 2021.
- Silver medal, Ibero-American Physics Olympiad, Colombia, 2017.
- Silver medal, Central-American and Carribean Physics Olympiad, El Salvador, 2016.

Conferences and Workshops attended

- Particle interactive systems: Analysis and computational methods, Berkeley, California, USA, June 2024.
- 33rd International Symposium on Rarefied Gas Dynamics, Göttingen, Germany, July 2024.
- SIAM Texas-Louisiana Sectional Meeting, Lafayette, Louisiana, USA, November 2023.
- SIAM Texas-Louisiana, Sectional Meeting, Waco, Texas, USA, October 2024.

Programming Languages

- C++, Matlab, Python, Java.

Tools

- $\LaTeX$, Linux, Git, Docker, Slurm.

Professional Experience

- **Research Experience**
 - Los Alamos National Laboratory, *Los Alamos, New Mexico*, **Graduate Research Intern**, 2025 - Present.
 - The University of Texas at Austin, *Austin, Texas*, **Research Assistant**, 2023 - Present.
 - Emory University, *Atlanta, Georgia*, **Research Assistant**, 2021 - 2022.
- **Teaching Experience**
 - The University of Texas at Austin, Austin, Texas, MATH 372K *Partial Differential Equations (PDEs) and Applications*, **Teaching Assistant**, Spring 2025.
 - Emory University, Atlanta, Georgia, PHYS-141 *Introduction to Physics 1*, **Tutor**, 2021 - 2022.
 - Emory University, Atlanta, Georgia, PHYS-142 *Introduction to Physics 2*, **Tutor**, 2021 - 2022.

References

- **Dr. Irene Gamba** Doctoral Advisor. gamba@math.utexas.edu
- **Dr. Diego Del Castillo-Negrete** Doctoral Co-Advisor. diego.delcastillo@austin.utexas.edu
- **Dr. William T. Taitano** Los Alamos National Laboratory, Staff Scientist. taitano@lanl.gov